

1 Data Preparation

- **TriviaQA** rc.nocontext
- 2,000 train / 200 eval questions
- **K₀** factual baseline (held-out)
- Bidirectional substring matching

2 Recursive Training

- **QLoRA** 4-bit NF4 / FFT
- Fresh adapter each generation
- **10 generations** $G_1 \rightarrow G_{10}$
- 3 backbones: Qwen, Gemma 3, 4

$\hookrightarrow$ Output Gen $t \rightarrow$ training for Gen $t+1$

3 Pressure Axes

- **Adapter rank** — 4, 16, 32, 64, 128, 256
- **Learning rate** — $1e-6$ to $2e-5$
- **Rank $\times$ LR** interaction matrix
- **Exposure** — 100% $\rightarrow$ ~95% synthetic

4 Evaluation

- **K₀ retention** — factual preservation
- **Distribution shift** — KL, JS, coverage
- **Behavioral** — persistence, length, repetition
- **Effective rank** — $\exp(H(\sigma))$ of $B \times A$

5 Regime Classification

Homeostatic — retention > 90%, stable

Bounded — stable retention, distribution drifts

Degradative — retention declines ~2-6 pp/gen

6 Pressure Framework

- **ETP threshold** identification
- Rank, LR, exposure $\rightarrow$ unified pressure
- Cross-backbone generalization
- Phase transition characterization